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Data-grounded discovery for SAP transformation

Why the assessment phase of an SAP engagement fails when it relies on interviews alone, and how modern tooling — applied to data the client already owns — moves the assessment forward in a structured, repeatable way.

Executive summary

Every SAP transformation programme opens with a discovery phase. In most engagements that phase produces a deliverable made of interview notes, sanitised workshop outputs, and an aspirational future-state map. The document rarely survives contact with the actual data. Once the design phase starts and the SI encounters the exceptions the business never mentioned in the workshop, the requirements begin to drift, and the programme absorbs the drift as change orders.

This paper argues that discovery deliverables should be grounded in the client's own data before they are grounded in interviews. Modern AI tooling — LLM-based document ingestion, structured-data reasoning, versioned prompt pipelines — has closed the gap between what a discovery team can process manually and what actually exists in the client's landscape. The tools now let a small specialist team read every contract, every exception log, every extract, and every prior programme's post-mortem, and arrive at the requirements conversation with a document the client can defend to a CFO.

None of this requires the client to buy a new AI product, license a new platform, or let data leave their own environment. The tooling runs on infrastructure the client already owns, and the client keeps custody of every derived artifact.

1. The interview-only discovery failure mode

The standard SAP discovery deliverable is a set of process maps, a list of pain points, and a target-state architecture, all synthesised from a fixed number of stakeholder interviews. This model is not wrong; it is incomplete. Its incompleteness has a specific shape: the interviews capture what the business thinks it does, not what the systems record it doing.

In practice this gap surfaces in three places:

Exception volume. The senior manager describes a clean process that handles 95% of transactions. The system logs show that 30% of transactions are exceptions handled by two people in a shared inbox. Neither the manager nor the two people know the volume; only the log knows.

Data-quality baselines. The workshop captures the intended data model. The actual data in the current system diverges from that model in ways nobody has characterised, because nobody has read every row.

Cross-system reconciliation gaps. Two adjacent systems report different totals for what should be the same figure. Everyone in the business knows a spreadsheet reconciles them. Nobody remembers who owns the spreadsheet.

A discovery deliverable that does not surface these three categories will produce a target-state design that cannot be built as specified. The design phase then discovers them and re-scopes; the programme absorbs the re-scoping as delay and cost.

2. What data-grounded discovery adds

A data-grounded discovery does everything the interview-only discovery does, and then adds a data pass. The data pass has a fixed shape:

2.1 · Document ingestion

Every contract, SOP, prior consulting deliverable, exception log, and known-issue register is ingested into a searchable index. Modern LLM-based document pipelines handle PDF, DOCX, XLSX, and image-only scans with OCR. The output is a corpus the specialist team can query in natural language.

2.2 · Structured extract

Sample extracts from the incumbent SAP or non-SAP systems — general ledger, sales order history, purchase order history, master data — are loaded into an ephemeral analysis environment. The environment is bounded, versioned, and destroyed at the end of the engagement; nothing is retained after the discovery phase ends.

2.3 · Reconciliation matrix

Where the workshop and the data disagree, the disagreement is written up as a numbered row in a reconciliation matrix. Each row cites the workshop statement, the data observation, the specialist's interpretation, and a recommended resolution path. The matrix is the discovery deliverable's core artifact.

2.4 · Requirements ratification

The workshop stakeholders reconvene with the reconciliation matrix on the screen. Every disagreement gets an owner and a decision. The requirements document that comes out of this session is defensible to the CFO, because every claim in it has a cited source in either the workshop record or the data extract.

3. What the tooling actually is

The stack that makes this practical is short and well-understood. The specific components matter less than the shape: an LLM-based reasoning surface, a bounded analytical database for extracts, and a versioned pipeline that traces every derived claim to a source document or a source row.

Two properties of the tooling are non-negotiable in a client engagement:

The tooling runs on infrastructure the client already owns. The reasoning surface sits in the client's tenant, on the client's compute. The client's IT security team reviews it against the same standards as any other application. Nothing crosses the client's boundary.

Every derived claim is traceable. The specialist team does not present conclusions the client cannot audit. Every number in the discovery deliverable is traceable to a source row; every summary is traceable to a source document. The AI does not add opinion; it accelerates the reading.

4. The economy for the client

The value of a data-grounded discovery is not measured in the discovery phase; it is measured in the design phase and in the change-order register that never gets written. In the engagements where this approach has replaced the interview-only pattern, the shape of the improvement is consistent:

Fewer surprises in design. The specialists who write the target-state design start from a requirements document that already accounts for the exception volume, the data-quality baseline, and the reconciliation gaps. Design cycles shorten.

Faster client decisions. The business sponsor can approve or reject scope items based on cited data instead of on a specialist's opinion. Sign-off cycles shorten.

A defensible position with SAP or the SI. When the client wants to push back on an SAP proposal or an SI's scope, they push back with cited evidence rather than with a counter-opinion. Commercial negotiations move faster.

5. When to engage this pattern

The pattern is worth the investment in three situations:

Pre-RFP. Before the client issues an RFP for the transformation programme, so the RFP itself is grounded and the vendor responses can be compared on a like-for-like basis.

Between vendor selection and contract signature. To pressure-test the winning vendor's proposed scope against the actual data before the commercial commitment is made.

Programme rescue. Mid-programme, when a workstream is stuck because the requirements are drifting and nobody has the specialist bench to read enough of the client's own material to converge.

6. About Steel Cloud Solutions

Steel Cloud Solutions is a boutique enterprise-software consultancy. Our SAP practice is client-side advisory — training, integration evaluation, project planning, and specialist technical staffing — for enterprise IT teams running or evaluating SAP. We are not an SAP channel partner, and we are not competing for SI scope. We work alongside your existing partners, on the specific questions where a specialist read of your own data would move the programme forward.

To scope a discovery engagement, email contact@steelcloudsolutions.com.